Termeh Taheri

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Profile

I work on audio understanding and multimodal perception, with a focus on building interpretable and user-friendly AI systems. I am interested in explainable NLP and in developing representations that connect audio and human language in more transparent and trustworthy ways. My recent work includes symbolic audio reasoning (SAR-LM) and evaluation tools that help users understand how generative models behave in real settings.

Experience

AI Engineer — London College of Business Studies
Jul 2025 - Dec 2025

- Built an AI-tutoring system that combines generative text models with controllable prompting and structured reasoning (CoT) to produce clear and reliable explanations for students.
- Integrated the AI components into the full software stack (backend and frontend), designing user-facing features that support understandable and interactive learning experiences.
- Deployed the system on AWS using FastAPI, Docker, and CI/CD pipelines, focusing on reproducibility, monitoring, and trustworthy model behaviour in real settings.

Software Engineer — Roshan
Aug 2022 - Aug 2024

- Refactored and optimized backend services (Django, FastAPI, PostgreSQL) to improve API performance, reliability, and maintainability.
- Enhanced data pipelines by redesigning preprocessing algorithms, improving their speed and consistency for downstream ML tasks.
- Worked on deployment and CI/CD processes, ensuring reproducible releases and stable integration of ML features into production systems.

Teaching Assistant, Data Structures — Noshirvani University of Technology
Feb 2020 - Jun 2020

- Guided students through core data structure concepts and supported them with their course projects.
- Held weekly help sessions to explain algorithms and troubleshoot implementation issues.
- Assisted with quizzes and exams, and graded assignments and final projects.

Publications

SAR-LM: Symbolic Audio Reasoning with Large Language Models (Paper) (Code)

- *LLM4MA Workshop @ ISMIR 2025; under review at ICASSP 2026.*
- Contribution: Initiated the project, developed the full reasoning pipeline, and implemented all experiments, with supervisory guidance on refining the approach and evaluation. Wrote the manuscript with supervisory feedback.

OmniVideoBench: Towards Audio-Visual Understanding Evaluation for Omni MLLMs (Paper)

- *Under review at ICLR 2026.*
- Contribution: Contributed to dataset development through audio-visual annotation and designed challenging test cases. Evaluated several state-of-the-art multimodal models to verify the benchmark's difficulty and diagnostic value.

Breast Cancer Prediction by Ensemble Meta-Feature Space Generator Based on Deep Neural Network (Paper)

- Biomedical Signal Processing and Control (2024).
- Contribution: Proposed the research idea, designed and implemented the full method and experiments, and wrote the manuscript with supervisory feedback.

Presentation of Encryption Method for RGB Images Based on an Evolutionary Algorithm Using Chaos Functions and Hash Tables (Paper)

- Multimedia Tools and Applications (2023).
- Contribution: Performed additional evaluation experiments to strengthen the method's analysis and wrote the full manuscript, incorporating supervisory feedback.

Selected Projects

CMI-Arena — Large-Scale Evaluation Platform for Text-to-Music Models

- Integrated text-to-music models as GPU-accelerated FastAPI services with efficient batching and streaming for scalable multimodal generation.
- Built preference-based evaluation pipelines to study how humans compare generative models in real settings.

SAR-LM — Symbolic Audio Reasoning with Large Language Models (Code)

- Built a modular PyTorch system for audio–text reasoning using symbolic features (PANNs, Whisper, MT3) and integrated Gemini/Qwen models for reproducible multimodal evaluation.
- Designed interpretable reasoning steps that reduce hallucinations and provide clear decision traces for understanding model behaviour.
- Evaluated the system on the MMAU, MMAR, and OmniBench benchmarks to study reasoning failures and support more transparent and trustworthy audio-language models.

Education

MSc Artificial Intelligence — Queen Mary University of London (Distinction) — September 2025

- *Thesis*: SAR-LM: Symbolic Audio Reasoning with Large Language Models
- *Advisor*: Prof. Emmanouil Benetos
- *Relevant modules:* Neural Networks & NLP; Natural Language Processing; Machine Learning; Neural Networks & Deep Learning; Artificial Intelligence; Ethics, Regulation and Law in Advanced Digital Information Processing; Information Retrieval; Applied Statistics.

BSc Computer Engineering — Noshirvani University of Technology — July 2022

- *Thesis*: Ensemble Deep Learning Algorithm for Medical Image Analysis
- *Relevant modules:* Data Mining, Data Structures, Algorithms

Software Proficiency

- **Machine Learning & NLP:** Python, PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn, JAX (basic), Keras.
- **Audio & Signal Processing:** Librosa, Essentia, FFmpeg, OpenCV.
- **Backend & Systems:** FastAPI, Django, Docker, Linux, PostgreSQL, Redis, Celery, ElasticSearch, Pytest.
- **Cloud & Deployment:** AWS (EC2), Dockerization, CI/CD workflows.
- **Other Programming:** Java, C++, MATLAB, React, HTML, CSS.

Honors & Awards

- **Chevening Scholarship —** Full scholarship recipient for MSc Artificial Intelligence, QMUL — 2024–2025
- **Best Bachelor Project Award —** First place across all departments, Noshirvani University of Technology — 2022
- **Research Grant (No P/M/1110) —** Supported publication in *Biomedical Signal Processing and Control* — 2021